RV University — School of Computer Science and Engineering (SoCSE)

Explainable AI (XAI) — Laboratory Record

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Subject: Explainable AI (XAI)	Experiment No.: 1

Ex No: 3 Date:	Local Interpretable Model-Agnostic Explanations (LIME)

Objective

To apply Local Interpretable Model-Agnostic Explanations (LIME) to a tabular machine-learning classification problem using the Adult/Census Income dataset, and to understand how individual feature conditions influence model predictions. The experiment also evaluates the underlying classifiers, explains correct and incorrect predictions, distinguishes local contribution from causation, and studies the stability of LIME explanations under repeated runs and different numbers of perturbation samples.

Description

In this laboratory, the Adult/Census Income dataset from the UCI Machine Learning Repository is used as a binary classification problem. The target variable, income, is encoded into two classes: 0 for <=50K and 1 for >50K. The dataset contains 48,842 instances and 14 input features consisting of six numerical and eight categorical attributes.

The workflow first examines the dataset and its data-quality issues. Numerical features are handled using median imputation and standardisation, while categorical features are imputed using the most frequent value and one-hot encoded. Because LimeTabularExplainer expects categorical variables in an integer-coded representation with category names, a separate LIME-compatible representation is constructed.

Two classifiers are trained: a Random Forest and Logistic Regression. Both are evaluated using accuracy, precision, recall, F1-score and ROC-AUC. Although Logistic Regression obtains a slightly higher F1-score, Random Forest is selected as the black-box model for LIME because its ensemble of decision trees is less directly interpretable than a coefficient-based linear model.

LIME is then used to generate local explanations for selected test instances, including correct predictions and a misclassification. Finally, explanation stability is assessed over repeated LIME runs and by changing num_samples from 1,000 to 5,000 and 10,000.

Model

1. Random Forest Classifier

The Random Forest classifier is an ensemble model composed of many decision trees. Each tree makes a prediction and the ensemble combines their outputs. This generally gives strong predictive performance, but the complete decision process can be difficult to inspect manually because many trees and interacting splits contribute to the final prediction. This makes Random Forest an appropriate black-box model for demonstrating a model-agnostic explanation method such as LIME.

2. Logistic Regression

Logistic Regression is trained as a comparison model. Its coefficients provide a relatively direct global interpretation, making it useful for comparing an inherently more transparent model with the black-box Random Forest.

3. LIME Tabular Explainer

LIME explains an individual prediction by generating perturbed samples around the selected instance, obtaining the black-box model's predictions for those samples, weighting the samples according to locality, and fitting a simple local surrogate model. The resulting LIME weights indicate which feature conditions support or oppose the class being explained for that particular instance.

Algorithm: LIME-Based Explainability Analysis

1. Start.
2. Import the required Python libraries and machine-learning algorithms.
3. Load the Adult/Census Income dataset from the UCI Machine Learning Repository.
4. Inspect the dataset shape, target variable, feature names and feature types.
5. Identify missing values, '?' placeholders, duplicate records and class imbalance.
6. Separate input features X and target variable y.
7. Encode income as 0 for <=50K and 1 for >50K.
8. Create a preprocessing pipeline using median imputation and StandardScaler for numerical features.
9. Create categorical preprocessing using most-frequent imputation and one-hot encoding.
10. Construct an integer-coded LIME-compatible representation for categorical variables.
11. Perform a stratified 80:20 train/test split using random seed 42.
12. Train a Random Forest classifier with 300 trees and balanced class weights.
13. Train a Logistic Regression classifier with balanced class weights.
14. Evaluate both models using accuracy, precision, recall, F1-score and ROC-AUC.
15. Compare the classifiers and select Random Forest as the black-box model for LIME.
16. Create a LimeTabularExplainer with the LIME-compatible feature representation and class names.
17. Select representative test instances and obtain their actual and predicted classes.
18. Generate local LIME explanations and record positive and negative feature contributions.
19. Interpret the contributions for each selected instance.
20. Identify a misclassified test instance and generate its LIME explanation.
21. Analyse whether the local evidence is consistent or conflicting.
22. Explain why LIME contributions do not establish causal relationships.
23. Repeat LIME explanations for the same instance to assess stability.
24. Calculate mean, standard deviation, minimum, maximum, top-five appearance count and coefficient of variation.
25. Repeat the stability experiment using 1,000, 5,000 and 10,000 perturbation samples.
26. Compare the resulting variability and interpret the effect of sample size.
27. Summarise the findings and limitations of local explanations.
28. Stop.

Results

Dataset Loading and Basic Information

Output

```
Loading Adult / Census Income dataset...
Dataset source: UCI Machine Learning Repository

Dataset loaded successfully!
Shape: (48842, 14)
```

	age	workclass	fnlwgt	education	education-num	marital-status	occupation	relationship	race	sex	capital-gain	capital-loss	hours-per-week	native-country
0	39	State-gov	77516	Bachelors	13	Never-married	Adm-clerical	Not-in-family	White	Male	2174	0	40	United-States
1	50	Self-emp-not-inc	83311	Bachelors	13	Married-civ-spouse	Exec-managerial	Husband	White	Male	0	0	13	United-States
2	38	Private	215646	HS-grad	9	Divorced	Handlers-cleaners	Not-in-family	White	Male	0	0	40	United-States
3	53	Private	234721	11th	7	Married-civ-spouse	Handlers-cleaners	Husband	Black	Male	0	0	40	United-States
4	28	Private	338409	Bachelors	13	Married-civ-spouse	Prof-specialty	Wife	Black	Female	0	0	40	Cuba

Figure 1: Adult/Census Income dataset loaded successfully and sample records displayed.

Output

```
Target variable: income

Target classes:
0    37155
1    11687
Name: count, dtype: int64
```

	age	workclass	fnlwgt	education	education_num	marital_status	occupation	relationship	race	sex	capital_gain	capital_loss	hours_per_week	native_country
0	39	State-gov	77516	Bachelors	13	Never-married	Adm-clerical	Not-in-family	White	Male	2174	0	40	United-States
1	50	Self-emp-not-inc	83311	Bachelors	13	Married-civ-spouse	Exec-managerial	Husband	White	Male	0	0	13	United-States
2	38	Private	215646	HS-grad	9	Divorced	Handlers-cleaners	Not-in-family	White	Male	0	0	40	United-States
3	53	Private	234721	11th	7	Married-civ-spouse	Handlers-cleaners	Husband	Black	Male	0	0	40	United-States
4	28	Private	338409	Bachelors	13	Married-civ-spouse	Prof-specialty	Wife	Black	Female	0	0	40	Cuba

Figure 2: Target variable and class counts displayed.

Output

```
============================================================
ADULT / CENSUS INCOME DATASET INFORMATION
============================================================
Number of instances : 48842
Number of features  : 14
Target variable     : income

Feature names:
 1. age
 2. workclass
 3. fnlwgt
 4. education
 5. education_num
 6. marital_status
 7. occupation
 8. relationship
 9. race
10. sex
11. capital_gain
12. capital_loss
13. hours_per_week
14. native_country

Numerical features:
['age', 'fnlwgt', 'education_num', 'capital_gain', 'capital_loss', 'hours_per_week']

Categorical features:
['workclass', 'education', 'marital_status', 'occupation', 'relationship', 'race', 'sex', 'native_c

Number of numerical features   : 6
Number of categorical features: 8

Target encoding:
0 = <=50K
1 = >50K
```

Figure 3: Dataset dimensions, feature names, numerical/categorical feature counts and target encoding.

Observation: The dataset was loaded successfully with 48,842 instances and 14 input features. The target is income, with class 0 representing <=50K and class 1 representing >50K.

Data Quality Analysis

Output

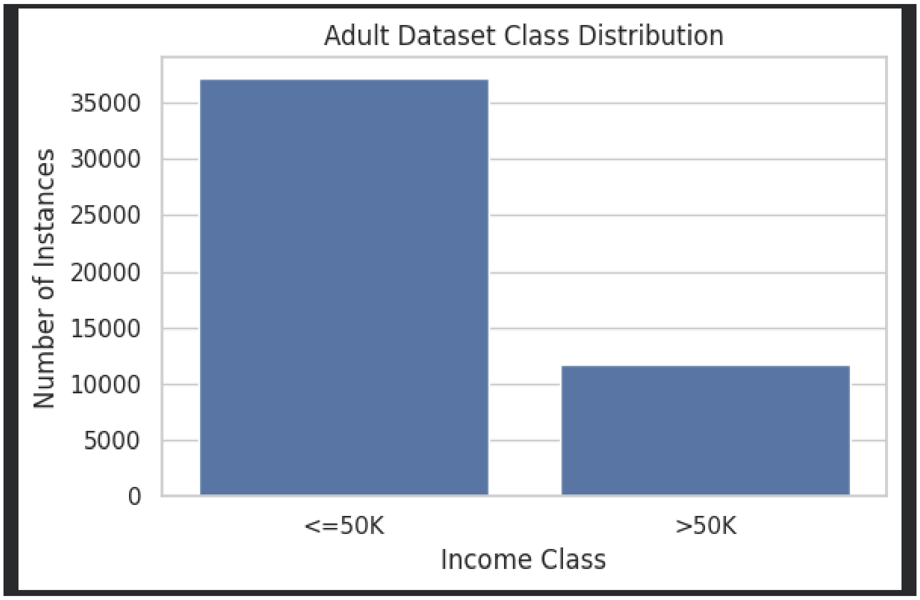


Figure 4: Data-quality checks showing missing values, '?' placeholders, duplicates and class distribution.

Output

```
Preprocessing pipeline created successfully.

Numerical preprocessing:
  1. Median imputation
  2. StandardScaler

Categorical preprocessing:
  1. Most-frequent imputation
  2. One-hot encoding
  3. Unknown categories ignored during transformation

Why categorical features require special handling in LIME:

LimeTabularExplainer expects categorical variables to be represented
using integer category codes together with categorical_names.

Therefore, LIME will use a separate integer-coded representation of
the original categorical variables. The trained machine-learning
model can still use one-hot encoded data internally.

This prevents LIME from treating categorical values such as
education categories as continuous numerical quantities.
```

Figure 5: Bar chart of the Adult dataset class distribution.

Observation: The dataset contains 2,203 explicit missing values and additional '?' placeholders in workclass, occupation and native_country. There are 53 duplicate records. The target is imbalanced: 37,155 records (76.07%) are <=50K and 11,687 (23.93%) are >50K.

Lab Exercises

Part A — Data Understanding and Preprocessing

Question 1

How many instances and features are present, and what are the feature types?

Answer: The dataset contains 48,842 instances and 14 input features. Six features are numerical: age, fnlwgt, education_num, capital_gain, capital_loss and hours_per_week. Eight are categorical: workclass, education, marital_status, occupation, relationship, race, sex and native_country. The target variable is income.

Output

```
============================================================
ADULT / CENSUS INCOME DATASET INFORMATION
============================================================
Number of instances : 48842
Number of features  : 14
Target variable     : income

Feature names:
 1. age
 2. workclass
 3. fnlwgt
 4. education
 5. education_num
 6. marital_status
 7. occupation
 8. relationship
 9. race
10. sex
11. capital_gain
12. capital_loss
13. hours_per_week
14. native_country

Numerical features:
['age', 'fnlwgt', 'education_num', 'capital_gain', 'capital_loss', 'hours_per_week']

Categorical features:
['workclass', 'education', 'marital_status', 'occupation', 'relationship', 'race', 'sex', 'native_c

Number of numerical features  : 6
Number of categorical features: 8

Target encoding:
0 = <=50K
1 = >50K
```

Figure 6: Dataset information output.

Observation: The dataset is mixed-type tabular data, so numerical and categorical variables require different preprocessing strategies.

Question 2

What data-quality issues were identified?

Answer: The experiment identified 2,203 explicit missing values, '?' placeholders in three categorical columns, and 53 duplicate records. The target distribution is imbalanced, with 76.07% of instances in the <=50K class and 23.93% in the >50K class.

Output

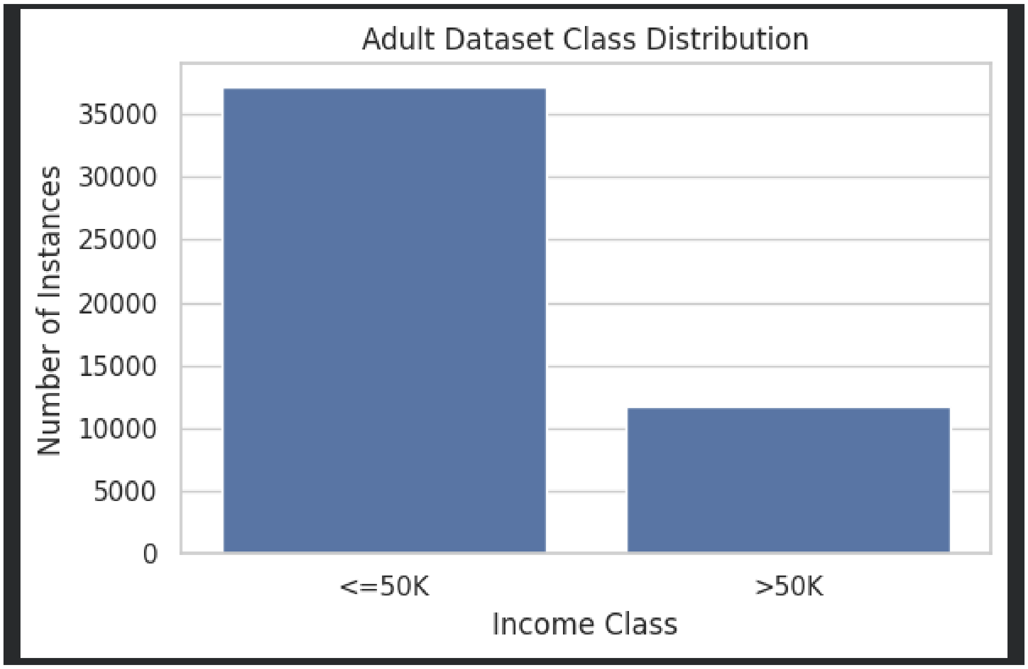


Figure 7: Data-quality check output.

Observation: Both missing values and '?' placeholders must be handled before model training. Class imbalance is also important when interpreting accuracy alone.

Question 3

How were the numerical and categorical features preprocessed?

Answer: Numerical features were processed using median imputation followed by StandardScaler. Categorical features were processed using most-frequent imputation followed by one-hot encoding. Unknown categories are ignored during transformation so that unseen categories do not cause transformation failure.

Output

```
LIME-compatible representation created.

Number of LIME features: 14

Categorical feature indices:
[1, 3, 5, 6, 7, 8, 9, 13]

Example categorical_names:
1 : ['Federal-gov', 'Local-gov', 'Never-worked', 'Private', 'Self-emp-inc', 'Self-emp-not-inc', 'State-gov', 'Without-pay', 'nan']
3 : ['10th', '11th', '12th', '1st-4th', '5th-6th', '7th-8th', '9th', 'Assoc-acdm', 'Assoc-voc', 'Bachelors']
5 : ['Divorced', 'Married-AF-spouse', 'Married-civ-spouse', 'Married-spouse-absent', 'Never-married', 'Separated', 'Widowed']
6 : ['Adm-clerical', 'Armed-Forces', 'Craft-repair', 'Exec-managerial', 'Farming-fishing', 'Handlers-cleaners', 'Machine-op-inspct', 'Other-service', 'Priv-house-serv', 'Prof-specialty']
7 : ['Husband', 'Not-in-family', 'Other-relative', 'Own-child', 'Unmarried', 'Wife']
```

	age	workclass	fnlwgt	education	education_num	marital_status	occupation	relationship	race	sex	capital_gain	capital_loss	hours_per_week	native_country
0	39	6	77516	9	13	4	0	1	4	1	2174	0	40	38
1	50	5	83311	9	13	2	3	0	4	1	0	0	13	38
2	38	3	215646	11	9	0	5	1	4	1	0	0	40	38
3	53	3	234721	1	7	2	5	0	2	1	0	0	40	38
4	28	3	338409	9	13	2	9	5	2	0	0	0	40	4

Figure 8: Preprocessing pipeline configuration.

Observation: The preprocessing pipeline separates numerical and categorical handling while producing a model-ready representation.

Question 4

Why do categorical features require special handling for LIME?

Answer: LimeTabularExplainer expects categorical variables to be represented using integer category codes together with categorical_names. If category codes were treated as ordinary continuous numbers, LIME could incorrectly interpret the numeric distance between category labels as meaningful. Therefore, the notebook constructs a separate LIME-compatible integer-coded representation, while the trained machine-learning model continues to use one-hot encoded categorical data.

Output

```
LIME-compatible representation created.

Number of LIME features: 14

Categorical feature indices:
[1, 3, 5, 6, 7, 8, 9, 13]

Example categorical_names:
1 : ['Federal-gov', 'Local-gov', 'Never-worked', 'Private', 'Self-emp-inc', 'Self-emp-not-inc', 'State-gov', 'Without-pay', 'nan']
3 : ['10th', '11th', '12th', '1st-4th', '5th-6th', '7th-8th', '9th', 'Assoc-acdm', 'Assoc-voc', 'Bachelors']
5 : ['Divorced', 'Married-AF-spouse', 'Married-civ-spouse', 'Married-spouse-absent', 'Never-married', 'Separated', 'Widowed']
6 : ['Adm-clerical', 'Armed-Forces', 'Craft-repair', 'Exec-managerial', 'Farming-fishing', 'Handlers-cleaners', 'Machine-op-inspct', 'Other-service', 'Priv-house-serv', 'Prof-specialty']
7 : ['Husband', 'Not-in-family', 'Other-relative', 'Own-child', 'Unmarried', 'Wife']
```

	age	workclass	fnlwgt	education	education_num	marital_status	occupation	relationship	race	sex	capital_gain	capital_loss	hours_per_week	native_country
0	39	6	77516	9	13	4	0	1	4	1	2174	0	40	38
1	50	5	83311	9	13	2	3	0	4	1	0	0	13	38
2	38	3	215646	11	9	0	5	1	4	1	0	0	40	38
3	53	3	234721	1	7	2	5	0	2	1	0	0	40	38
4	28	3	338409	9	13	2	9	5	2	0	0	0	40	4

Figure 9: Explanation of categorical preprocessing and LIME handling.

Output

```
Training samples: 39073
Testing samples : 9769

Training class distribution:
0    0.76073
1    0.23927
Name: proportion, dtype: float64

Testing class distribution:
0    0.760672
1    0.239328
Name: proportion, dtype: float64
```

Figure 10: LIME-compatible representation with categorical feature indices and category names.

Observation: Separate representations allow the model and LIME to use formats appropriate to their respective requirements without treating categorical labels as continuous quantities.

Part B — Model Training and Evaluation

Question 5

How was the train/test split performed?

Answer: A stratified 80:20 train/test split was used with random seed 42. The training set contains 39,073 samples and the testing set contains 9,769 samples. Stratification preserved approximately the same class proportions in both subsets.

Output

```
============================================================
Random Forest
============================================================

Classification Report:
              precision    recall  f1-score   support

       <=50K       0.88      0.93      0.91      7431
        >50K       0.73      0.62      0.67      2338

    accuracy                           0.85      9769
   macro avg       0.81      0.77      0.79      9769
weighted avg       0.85      0.85      0.85      9769

Confusion Matrix:
[[6906  525]
 [ 898 1440]]

============================================================
Logistic Regression
============================================================

Classification Report:
              precision    recall  f1-score   support

       <=50K       0.94      0.80      0.86      7431
        >50K       0.57      0.84      0.68      2338

    accuracy                           0.81      9769
   macro avg       0.75      0.82      0.77      9769
weighted avg       0.85      0.81      0.82      9769

Confusion Matrix:
[[5937 1494]
 [ 383 1955]]

Overall Model Performance:
```

	Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
0	Random Forest	0.8543	0.7328	0.6159	0.6693	0.9044
1	Logistic Regression	0.8079	0.5668	0.8362	0.6757	0.9044

Figure 11: Training and testing sample counts and class proportions.

Observation: The similar class proportions in training and testing sets indicate that stratification was successful.

Question 6

Which classification models were trained?

Answer: Two classifiers were trained. The Random Forest uses 300 trees, balanced class weights and random_state=42. Logistic Regression uses balanced class weights and max_iter=2000. Random Forest is subsequently selected as the black-box model for LIME.

Output

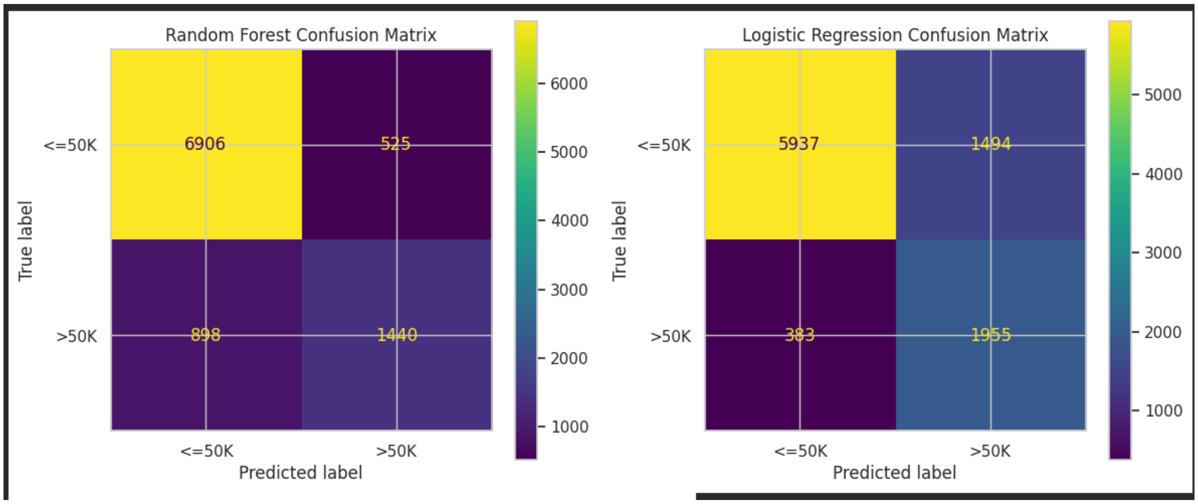


Figure 12: Model performance metrics for Random Forest and Logistic Regression.

Observation: Both models were trained on the same preprocessed data, allowing a direct performance comparison.

Question 7

What were the evaluation results?

Answer:

Model	**Accuracy**	**Precision**	**Recall**	**F1-score**	**ROC-AUC**
Random Forest	0.8543	0.7328	0.6159	0.6693	0.9044
Logistic Regression	0.8079	0.5668	0.8362	0.6757	0.9044

Output

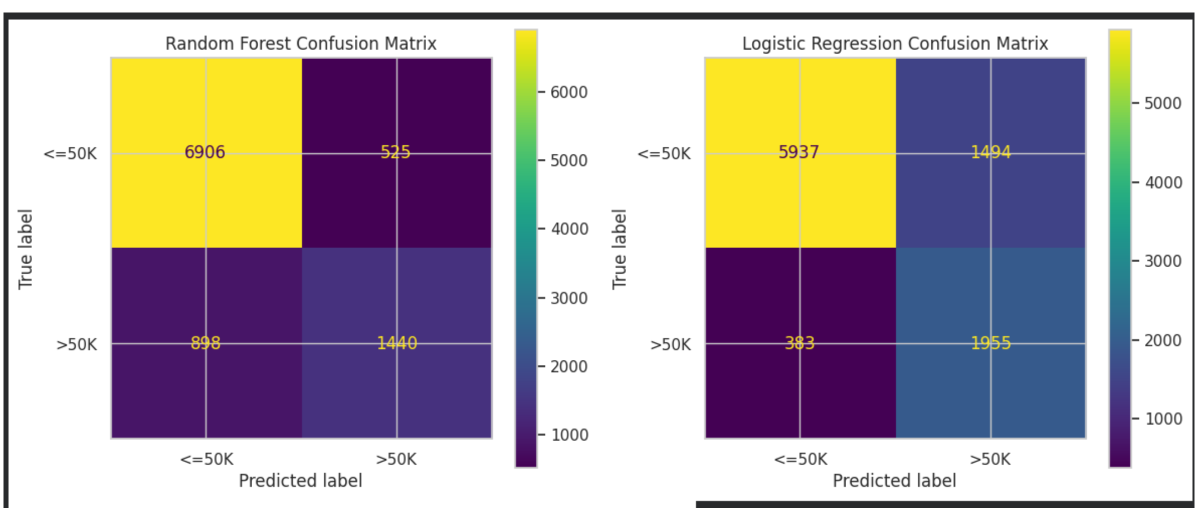


Figure 13: Classification reports and overall model performance table.

Output

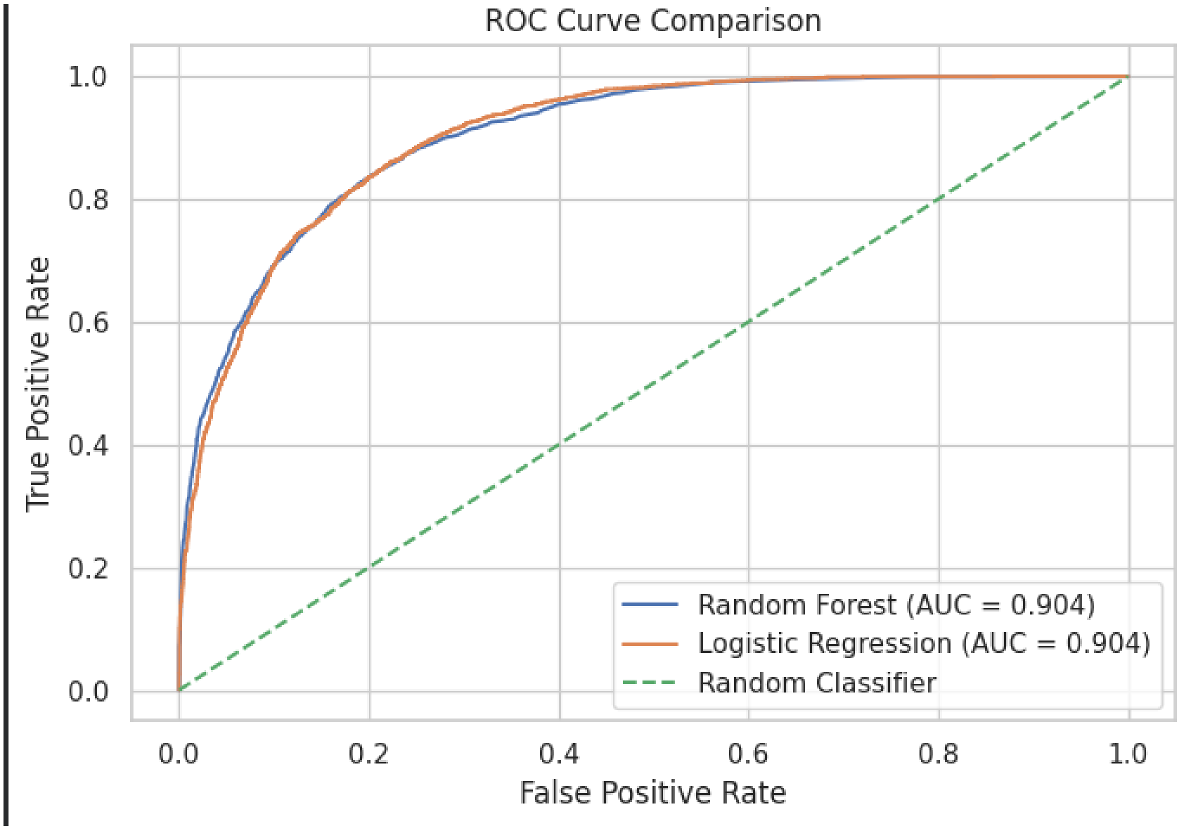


Figure 14: Confusion matrices for Random Forest and Logistic Regression.

Output

```
==============================================================
MODEL COMPARISON
==============================================================
```

	Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
0	Random Forest	0.8543	0.7328	0.6159	0.6693	0.9044
1	Logistic Regression	0.8079	0.5668	0.8362	0.6757	0.9044

```
Model with highest F1-score: Logistic Regression

Black-box model selected for LIME: Random Forest

Reason:
Random Forest consists of many decision trees and combines their
predictions through an ensemble process. Its decision-making process
is therefore less directly interpretable than Logistic Regression,
whose coefficients provide a more straightforward global interpretation.

LIME is model-agnostic and can explain the Random Forest using only
its prediction function.
```

Figure 15: ROC curve comparison.

Observation: Random Forest provides higher accuracy and precision, while Logistic Regression provides higher recall and a slightly higher F1-score. Both models have the same ROC-AUC of 0.9044. The confusion matrices show that Random Forest correctly classifies 6,906 <=50K and 1,440 >50K cases.

Question 8

Which model performed better overall?

Answer: Random Forest achieved higher accuracy (0.8543 versus 0.8079) and higher precision (0.7328 versus 0.5668), while Logistic Regression achieved a slightly higher F1-score (0.6757 versus 0.6693) and much higher recall for the >50K class (0.8362 versus 0.6159). Both models achieved ROC-AUC = 0.9044. For this laboratory, Random Forest is selected for LIME because it is less directly interpretable than Logistic Regression and therefore provides a more meaningful black-box explainability case.

Output

```
LimeTabularExplainer created successfully!

feature_names:
['age', 'workclass', 'fnlwgt', 'education', 'education_num', 'marital_status', 'occupation', 'relat

class_names:
['<=50K', '>50K']

categorical_features:
[1, 3, 5, 6, 7, 8, 9, 13]

categorical_names:
{'workclass': ['Federal-gov', 'Local-gov', 'Never-worked', 'Private', 'Self-emp-inc'], 'education':
```

Figure 16: Model comparison and black-box model selection.

Observation: Model selection is based not only on one metric but also on the interpretability purpose of the experiment.

Part C — LIME Explanation of Individual Predictions

Question 9

Why was Random Forest selected as the black-box model for LIME?

Answer: Random Forest combines many decision trees into an ensemble, making the complete decision process difficult to inspect manually. Logistic Regression, in contrast, has directly inspectable coefficients. Since LIME is model-agnostic and only needs access to a prediction function, it can explain the Random Forest without requiring access to its internal decision rules.

Output

```
LimeTabularExplainer created successfully!

feature_names:
['age', 'workclass', 'fnlwgt', 'education', 'education_num', 'marital_status', 'occupation', 'relat

class_names:
['<=50K', '>50K']

categorical_features:
[1, 3, 5, 6, 7, 8, 9, 13]

categorical_names:
{'workclass': ['Federal-gov', 'Local-gov', 'Never-worked', 'Private', 'Self-emp-inc'], 'education':
```

Figure 17: Black-box model selection output.

Question 10

How was the LIME explainer configured?

Answer: A LimeTabularExplainer was created using the 14-feature LIME-compatible representation. The class names are <=50K and >50K, and the categorical feature indices are supplied together with category names.

Output

```
Selected test instance indices: [np.int64(0), np.int64(4), np.int64(711)]
```

	Instance	Actual Class	Predicted Class	P(>50K)
0	0	<=50K	<=50K	0.0
1	4	>50K	>50K	0.6
2	711	>50K	<=50K	0.5

Figure 18: LimeTabularExplainer configuration.

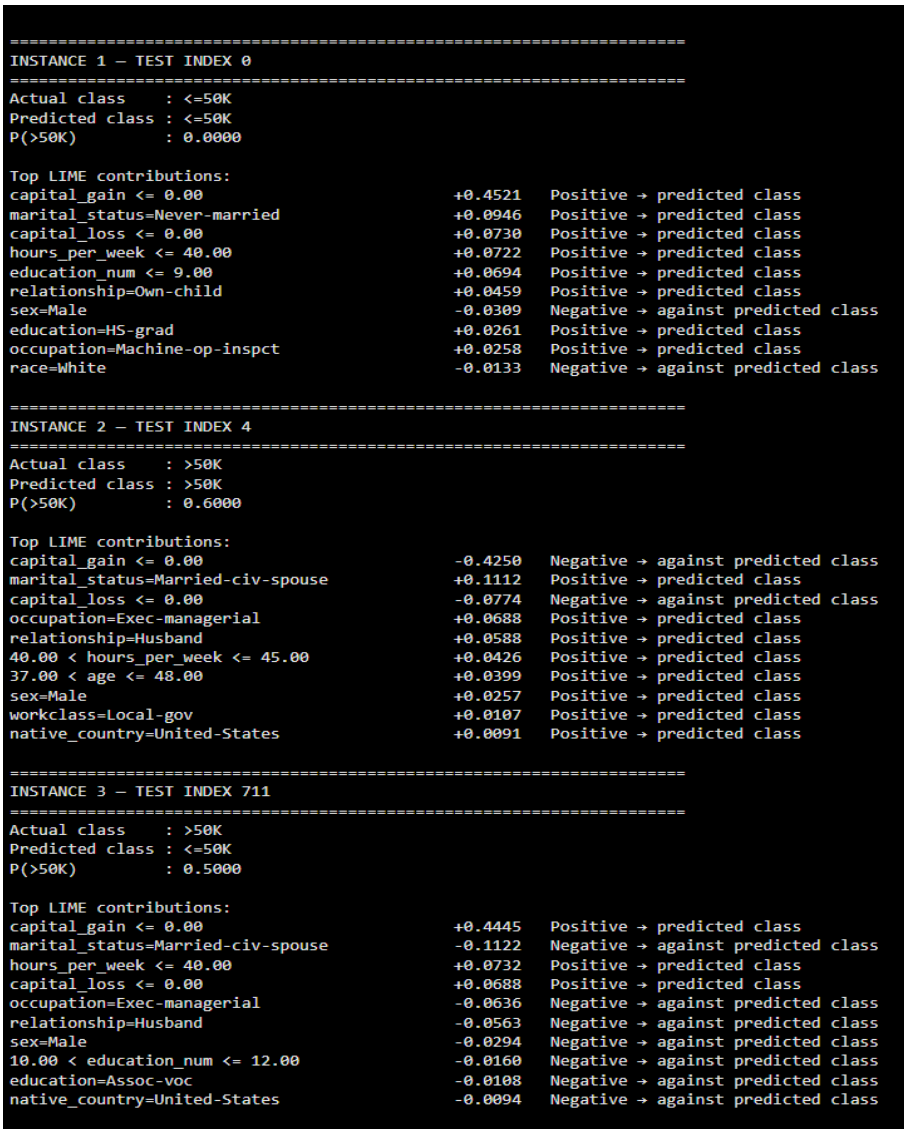
Observation: The LIME explainer is configured to recognise categorical variables explicitly, preventing category codes from being interpreted as continuous measurements.

Question 11

Which test instances were selected for local explanation?

Answer: Three representative test instances were selected: test index 0, test index 4 and test index 711. Their recorded outcomes were respectively <=50K predicted as <=50K with P(>50K)=0.0000; >50K predicted as >50K with P(>50K)=0.6000; and >50K predicted as <=50K with P(>50K)=0.5000.

Output



```
======================================================================
INSTANCE 1 — TEST INDEX 0
======================================================================
Actual class    : <=50K
Predicted class : <=50K
P(>50K)         : 0.0000

Top LIME contributions:
capital_gain <= 0.00                          +0.4521   Positive → predicted class
marital_status=Never-married                  +0.0946   Positive → predicted class
capital_loss <= 0.00                          +0.0730   Positive → predicted class
hours_per_week <= 40.00                       +0.0722   Positive → predicted class
education_num <= 9.00                         +0.0694   Positive → predicted class
relationship=Own-child                        +0.0459   Positive → predicted class
sex=Male                                      -0.0309   Negative → against predicted class
education=HS-grad                             +0.0261   Positive → predicted class
occupation=Machine-op-inspct                  +0.0258   Positive → predicted class
race=White                                    -0.0133   Negative → against predicted class

======================================================================
INSTANCE 2 — TEST INDEX 4
======================================================================
Actual class    : >50K
Predicted class : >50K
P(>50K)         : 0.6000

Top LIME contributions:
capital_gain <= 0.00                          -0.4250   Negative → against predicted class
marital_status=Married-civ-spouse             +0.1112   Positive → predicted class
capital_loss <= 0.00                          -0.0774   Negative → against predicted class
occupation=Exec-managerial                    +0.0688   Positive → predicted class
relationship=Husband                          +0.0588   Positive → predicted class
40.00 < hours_per_week <= 45.00               +0.0426   Positive → predicted class
37.00 < age <= 48.00                          +0.0399   Positive → predicted class
sex=Male                                      +0.0257   Positive → predicted class
workclass=Local-gov                           +0.0107   Positive → predicted class
native_country=United-States                  +0.0091   Positive → predicted class

======================================================================
INSTANCE 3 — TEST INDEX 711
======================================================================
Actual class    : >50K
Predicted class : <=50K
P(>50K)         : 0.5000

Top LIME contributions:
capital_gain <= 0.00                          +0.4445   Positive → predicted class
marital_status=Married-civ-spouse             -0.1122   Negative → against predicted class
hours_per_week <= 40.00                       +0.0732   Positive → predicted class
capital_loss <= 0.00                          +0.0688   Positive → predicted class
occupation=Exec-managerial                    -0.0636   Negative → against predicted class
relationship=Husband                          -0.0563   Negative → against predicted class
sex=Male                                      -0.0294   Negative → against predicted class
10.00 < education_num <= 12.00                -0.0160   Negative → against predicted class
education=Assoc-voc                           -0.0108   Negative → against predicted class
native_country=United-States                  -0.0094   Negative → against predicted class
```

Figure 19: Selected test instance indices and predictions.

Observation: The three cases cover a correct low-income prediction, a correct high-income prediction and a misclassification, allowing different local behaviours to be compared.

Question 12

What were the actual feature values of the selected instances?

Answer: Instance 1 is a 30-year-old Private-sector worker with HS-grad education, education_num 9, Never-married status, Machine-op-inspct occupation, Own-child relationship, zero capital gain/loss and 40 hours per week. Instance 2 is age 42, Local-gov, Some-college, education_num 10, Married-civ-spouse, Exec-managerial, Husband, zero capital gain/loss and 45 hours per week. Instance 3 is age 29, Private, Assoc-voc, education_num 11, Married-civ-spouse, Exec-managerial, Husband, zero capital gain/loss and 40 hours per week.*2.*

Output

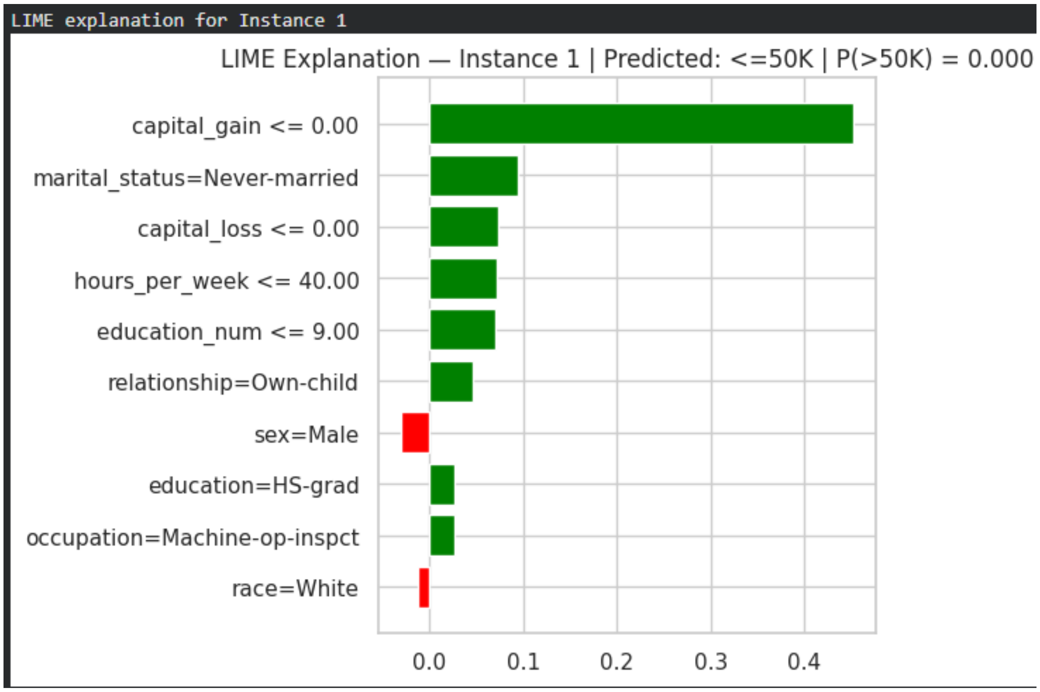


Figure 21: Actual feature values for Instance 3.

Question 13

What did LIME show for Instance 1?

Answer: Instance 1 was predicted as <=50K with P(>50K)=0.0000. The strongest positive contributor was capital_gain <=0.00 with weight +0.4521, followed by marital_status=Never-married (+0.0946), capital_loss <=0.00 (+0.0730), hours_per_week <=40.00 (+0.0722) and education_num <=9.00 (+0.0694). Negative contributors included sex=Male (-0.0309) and race=White (-0.0133).

Output

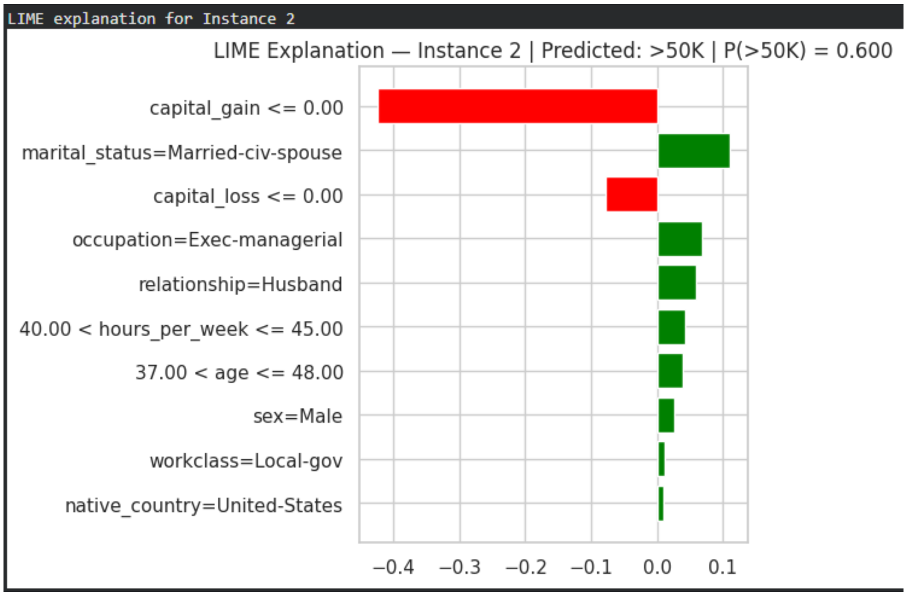


Figure 22: LIME bar-chart explanation for Instance 1.

Output

```
================================================================================
INTERPRETATION — INSTANCE 1
================================================================================
Predicted class: <=50K
Prediction probability P(>50K): 0.0000

Feature interpretation:

Feature condition: capital_gain <= 0.00
LIME weight: 0.4521
Direction: supports the predicted class
Effect: increases the local likelihood of the predicted class

Feature condition: marital_status=Never-married
LIME weight: 0.0946
Direction: supports the predicted class
Effect: increases the local likelihood of the predicted class

Feature condition: capital_loss <= 0.00
LIME weight: 0.073
Direction: supports the predicted class
Effect: increases the local likelihood of the predicted class

Feature condition: hours_per_week <= 40.00
LIME weight: 0.0722
Direction: supports the predicted class
Effect: increases the local likelihood of the predicted class

Feature condition: education_num <= 9.00
LIME weight: 0.0694
Direction: supports the predicted class
Effect: increases the local likelihood of the predicted class

Feature condition: sex=Male
LIME weight: -0.0309
Direction: opposes the predicted class
Effect: reduces the local likelihood of the predicted class

Feature condition: race=White
LIME weight: -0.0133
Direction: opposes the predicted class
Effect: reduces the local likelihood of the predicted class
```

Figure 23: Top positive and negative contributors for Instance 1.

Observation: The local explanation shows that several feature conditions support the <=50K prediction, with capital_gain <=0.00 dominating the local contribution.

Question 14

What did LIME show for Instance 2?

Answer: Instance 2 was predicted as >50K with P(>50K)=0.6000. Positive contributors included marital_status=Married-civ-spouse (+0.1112), occupation=Exec-managerial (+0.0688), relationship=Husband (+0.0588), 40<hours_per_week<=45 (+0.0426) and 37<age<=48 (+0.0399). The strongest negative contributor was capital_gain <=0.00 (-0.4250), followed by capital_loss <=0.00 (-0.0774).

Output

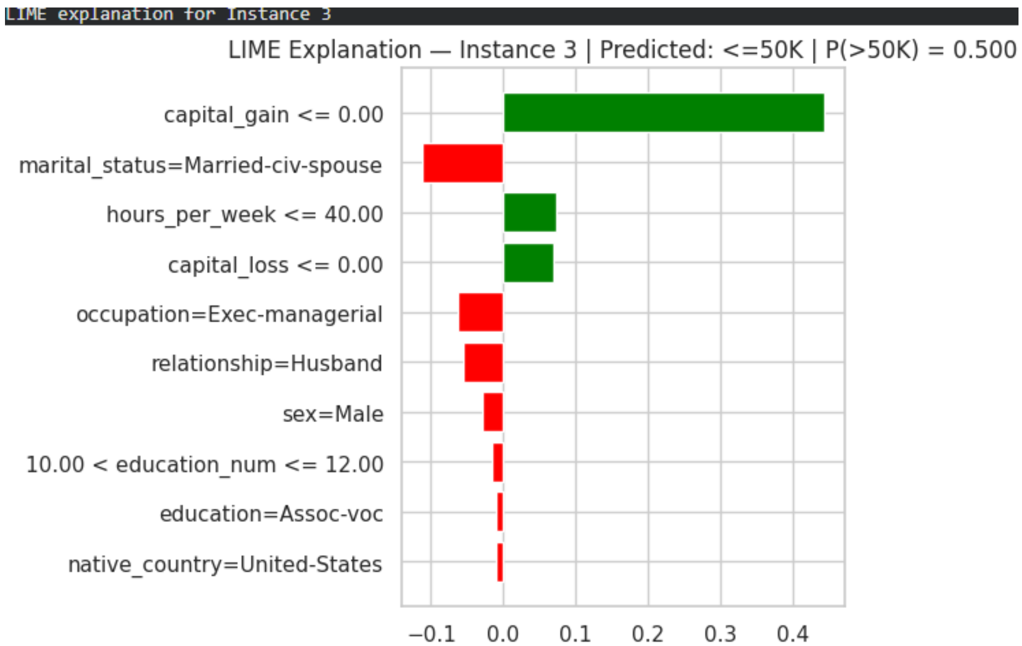


Figure 24: LIME bar-chart explanation for Instance 2.

Output

```
==========================================================================
INTERPRETATION — INSTANCE 1
==========================================================================
Predicted class: <=50K
Prediction probability P(>50K): 0.0000

Feature interpretation:

Feature condition: capital_gain <= 0.00
LIME weight: 0.4521
Direction: supports the predicted class
Effect: increases the local likelihood of the predicted class

Feature condition: marital_status=Never-married
LIME weight: 0.0946
Direction: supports the predicted class
Effect: increases the local likelihood of the predicted class

Feature condition: capital_loss <= 0.00
LIME weight: 0.073
Direction: supports the predicted class
Effect: increases the local likelihood of the predicted class

Feature condition: hours_per_week <= 40.00
LIME weight: 0.0722
Direction: supports the predicted class
Effect: increases the local likelihood of the predicted class

Feature condition: education_num <= 9.00
LIME weight: 0.0694
Direction: supports the predicted class
Effect: increases the local likelihood of the predicted class

Feature condition: sex=Male
LIME weight: -0.0309
Direction: opposes the predicted class
Effect: reduces the local likelihood of the predicted class

Feature condition: race=White
LIME weight: -0.0133
Direction: opposes the predicted class
Effect: reduces the local likelihood of the predicted class
```

Figure 25: Positive and negative contributor summary including Instance 2.

Observation: The same feature condition, capital_gain <=0, can oppose the predicted >50K class in this local explanation, demonstrating that LIME weights are class- and instance-dependent.

Question 15

What did LIME show for Instance 3?

Answer: Instance 3 was predicted as <=50K with P(>50K)=0.5000, although its actual class was >50K. Positive contributors toward the predicted class included capital_gain <=0.00 (+0.4445), hours_per_week <=40.00 (+0.0732) and capital_loss <=0.00 (+0.0688). Negative contributors included marital_status=Married-civ-spouse (-0.1122), occupation=Exec-managerial (-0.0636), relationship=Husband (-0.0563), sex=Male (-0.0294) and 10<education_num<=12 (-0.0160).

Output

```
==========================================================================
INSTANCE 1
Predicted class: <=50K
==========================================================================

TOP FIVE POSITIVE CONTRIBUTORS
1. capital_gain <= 0.00 | LIME weight = +0.4521
2. marital_status=Never-married | LIME weight = +0.0946
3. capital_loss <= 0.00 | LIME weight = +0.0730
4. hours_per_week <= 40.00 | LIME weight = +0.0722
5. education_num <= 9.00 | LIME weight = +0.0694

TOP FIVE NEGATIVE CONTRIBUTORS
1. sex=Male | LIME weight = -0.0309
2. race=White | LIME weight = -0.0133

==========================================================================
INSTANCE 2
Predicted class: >50K
==========================================================================

TOP FIVE POSITIVE CONTRIBUTORS
1. marital_status=Married-civ-spouse | LIME weight = +0.1112
2. occupation=Exec-managerial | LIME weight = +0.0688
3. relationship=Husband | LIME weight = +0.0588
4. 40.00 < hours_per_week <= 45.00 | LIME weight = +0.0426
5. 37.00 < age <= 48.00 | LIME weight = +0.0399

TOP FIVE NEGATIVE CONTRIBUTORS
1. capital_gain <= 0.00 | LIME weight = -0.4250
2. capital_loss <= 0.00 | LIME weight = -0.0774

==========================================================================
INSTANCE 3
Predicted class: <=50K
==========================================================================

TOP FIVE POSITIVE CONTRIBUTORS
1. capital_gain <= 0.00 | LIME weight = +0.4445
2. hours_per_week <= 40.00 | LIME weight = +0.0732
3. capital_loss <= 0.00 | LIME weight = +0.0688

TOP FIVE NEGATIVE CONTRIBUTORS
1. marital_status=Married-civ-spouse | LIME weight = -0.1122
2. occupation=Exec-managerial | LIME weight = -0.0636
3. relationship=Husband | LIME weight = -0.0563
4. sex=Male | LIME weight = -0.0294
5. 10.00 < education_num <= 12.00 | LIME weight = -0.0160
```

Figure 26: LIME bar-chart explanation for Instance 3.

Output

```
==========================================================================
INTERPRETATION — INSTANCE 1
==========================================================================
Predicted class: <=50K
Prediction probability P(>50K): 0.0000

Feature interpretation:

Feature condition: capital_gain <= 0.00
LIME weight: 0.4521
Direction: supports the predicted class
Effect: increases the local likelihood of the predicted class

Feature condition: marital_status=Never-married
LIME weight: 0.0946
Direction: supports the predicted class
Effect: increases the local likelihood of the predicted class

Feature condition: capital_loss <= 0.00
LIME weight: 0.073
Direction: supports the predicted class
Effect: increases the local likelihood of the predicted class

Feature condition: hours_per_week <= 40.00
LIME weight: 0.0722
Direction: supports the predicted class
Effect: increases the local likelihood of the predicted class

Feature condition: education_num <= 9.00
LIME weight: 0.0694
Direction: supports the predicted class
Effect: increases the local likelihood of the predicted class

Feature condition: sex=Male
LIME weight: -0.0309
Direction: opposes the predicted class
Effect: reduces the local likelihood of the predicted class

Feature condition: race=White
LIME weight: -0.0133
Direction: opposes the predicted class
Effect: reduces the local likelihood of the predicted class
```

Figure 27: Positive and negative contributor summary including Instance 1.

Observation: The explanation contains evidence in both directions. This is useful for understanding why the model's local decision was uncertain and ultimately incorrect.

Part D — Interpretation of LIME Contributions

Question 16

What are the five strongest positive contributors for Instance 1?

Answer: capital_gain <=0.00 (+0.4521), marital_status=Never-married (+0.0946), capital_loss <=0.00 (+0.0730), hours_per_week <=40.00 (+0.0722), and education_num <=9.00 (+0.0694).

Output

```
==========================================================================
INTERPRETATION — INSTANCE 1
==========================================================================
Predicted class: <=50K
Prediction probability P(>50K): 0.0000

Feature interpretation:

Feature condition: capital_gain <= 0.00
LIME weight: 0.4521
Direction: supports the predicted class
Effect: increases the local likelihood of the predicted class

Feature condition: marital_status=Never-married
LIME weight: 0.0946
Direction: supports the predicted class
Effect: increases the local likelihood of the predicted class

Feature condition: capital_loss <= 0.00
LIME weight: 0.073
Direction: supports the predicted class
Effect: increases the local likelihood of the predicted class

Feature condition: hours_per_week <= 40.00
LIME weight: 0.0722
Direction: supports the predicted class
Effect: increases the local likelihood of the predicted class

Feature condition: education_num <= 9.00
LIME weight: 0.0694
Direction: supports the predicted class
Effect: increases the local likelihood of the predicted class

Feature condition: sex=Male
LIME weight: -0.0309
Direction: opposes the predicted class
Effect: reduces the local likelihood of the predicted class

Feature condition: race=White
LIME weight: -0.0133
Direction: opposes the predicted class
Effect: reduces the local likelihood of the predicted class
```

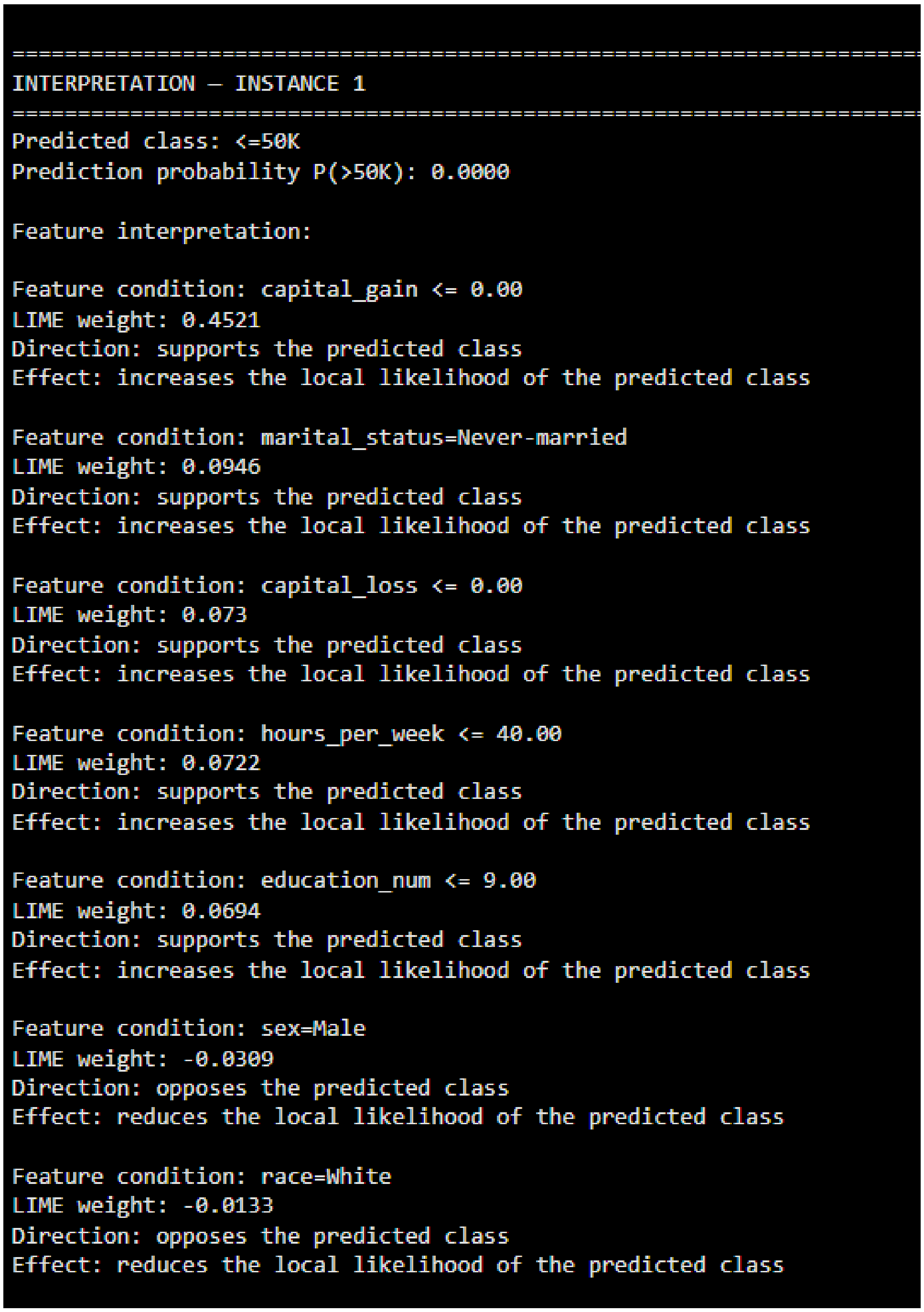
Figure 28: Top five positive and negative LIME contributors.

Question 17

What are the main negative contributors for the selected examples?

Answer: For Instance 1, sex=Male and race=White oppose the predicted class. For Instance 2, capital_gain <=0.00 and capital_loss <=0.00 oppose the >50K prediction. For Instance 3, married-civ-spouse, Exec-managerial, Husband, Male and the 10–12 education_num interval oppose the predicted <=50K class.

Output



```
================================================================
INTERPRETATION – INSTANCE 1
================================================================
Predicted class: <=50K
Prediction probability P(>50K): 0.0000

Feature interpretation:

Feature condition: capital_gain <= 0.00
LIME weight: 0.4521
Direction: supports the predicted class
Effect: increases the local likelihood of the predicted class

Feature condition: marital_status=Never-married
LIME weight: 0.0946
Direction: supports the predicted class
Effect: increases the local likelihood of the predicted class

Feature condition: capital_loss <= 0.00
LIME weight: 0.073
Direction: supports the predicted class
Effect: increases the local likelihood of the predicted class

Feature condition: hours_per_week <= 40.00
LIME weight: 0.0722
Direction: supports the predicted class
Effect: increases the local likelihood of the predicted class

Feature condition: education_num <= 9.00
LIME weight: 0.0694
Direction: supports the predicted class
Effect: increases the local likelihood of the predicted class

Feature condition: sex=Male
LIME weight: -0.0309
Direction: opposes the predicted class
Effect: reduces the local likelihood of the predicted class

Feature condition: race=White
LIME weight: -0.0133
Direction: opposes the predicted class
Effect: reduces the local likelihood of the predicted class
```

Figure 29: Positive and negative contributor summary.

Observation: Positive and negative weights should be read relative to the class being explained rather than as universal statements about income.

Question 18

How should the LIME contributions be interpreted?

Answer: A positive LIME weight supports the class currently being explained, while a negative weight opposes that class. The weight describes a local contribution to the model's prediction for the selected instance. It should not be interpreted as a global feature effect or a causal relationship.

Output

```
INTERPRETATION — INSTANCE 2
======================================================================
Predicted class: >50K
Prediction probability P(>50K): 0.6000

Feature interpretation:

Feature condition: marital_status=Married-civ-spouse
LIME weight: 0.1112
Direction: supports the predicted class
Effect: increases the local likelihood of the predicted class

Feature condition: occupation=Exec-managerial
LIME weight: 0.0688
Direction: supports the predicted class
Effect: increases the local likelihood of the predicted class

Feature condition: relationship=Husband
LIME weight: 0.0588
Direction: supports the predicted class
Effect: increases the local likelihood of the predicted class

Feature condition: 40.00 < hours_per_week <= 45.00
LIME weight: 0.0426
Direction: supports the predicted class
Effect: increases the local likelihood of the predicted class

Feature condition: 37.00 < age <= 48.00
LIME weight: 0.0399
Direction: supports the predicted class
Effect: increases the local likelihood of the predicted class

Feature condition: capital_gain <= 0.00
LIME weight: -0.425
Direction: opposes the predicted class
Effect: reduces the local likelihood of the predicted class

Feature condition: capital_loss <= 0.00
LIME weight: -0.0774
Direction: opposes the predicted class
Effect: reduces the local likelihood of the predicted class

----------------------------------------------------------------------
```

Figure 30: Detailed interpretation of LIME contributions for Instance 1.

Output

```
================================================================
INTERPRETATION — INSTANCE 3
================================================================
Predicted class: <=50K
Prediction probability P(>50K): 0.5000

Feature interpretation:

Feature condition: capital_gain <= 0.00
LIME weight: 0.4445
Direction: supports the predicted class
Effect: increases the local likelihood of the predicted class

Feature condition: hours_per_week <= 40.00
LIME weight: 0.0732
Direction: supports the predicted class
Effect: increases the local likelihood of the predicted class

Feature condition: capital_loss <= 0.00
LIME weight: 0.0688
Direction: supports the predicted class
Effect: increases the local likelihood of the predicted class

Feature condition: marital_status=Married-civ-spouse
LIME weight: -0.1122
Direction: opposes the predicted class
Effect: reduces the local likelihood of the predicted class

Feature condition: occupation=Exec-managerial
LIME weight: -0.0636
Direction: opposes the predicted class
Effect: reduces the local likelihood of the predicted class

Feature condition: relationship=Husband
LIME weight: -0.0563
Direction: opposes the predicted class
Effect: reduces the local likelihood of the predicted class

Feature condition: sex=Male
LIME weight: -0.0294
Direction: opposes the predicted class
Effect: reduces the local likelihood of the predicted class

Feature condition: 10.00 < education_num <= 12.00
LIME weight: -0.016
Direction: opposes the predicted class
Effect: reduces the local likelihood of the predicted class
```

Figure 31: Detailed interpretation of LIME contributions for Instance 3.

Output

```
Q19 — Interpretation of a Positive LIME Weight
================================================

If LIME assigns a positive weight to:

    education = Bachelors

this means that, for the particular individual being explained,
the presence of the Bachelor's education category contributes
towards the class currently being explained by LIME.

If the predicted class is >50K, the positive weight means that
education = Bachelors supports the model's local prediction of
income above $50K.

If the predicted class is <=50K, the positive weight means that
education = Bachelors supports the model's local prediction of
income at or below $50K.

Therefore, a positive LIME weight does NOT universally mean
"high income". Its meaning depends on which class is being
explained.

The LIME weight represents a local contribution to the model's
prediction; it does not prove that education causes income to
increase.
```

Figure 32: Detailed interpretation of LIME contributions for Instance 3.

Question 19

What does a positive LIME weight for education=Bachelors mean?

Answer: A positive weight means that, for the individual being explained, the condition education=Bachelors contributes toward the class currently being explained. If the explained class is >50K, the positive weight supports the local >50K prediction; if the explained class is <=50K, it supports the local <=50K prediction. Therefore, a positive weight does not universally mean high income, and it does not prove that education causes income to increase.

Output

	Concept	Meaning	Scope	Example
0	Feature Importance	Measures how important a feature is to the mod...	Usually global	Age may be highly important across the dataset.
1	LIME Weight	Measures how a feature condition contributes t...	Local / individual instance	For one person, age > 35 may push the predicti...
2	Prediction Probability	Measures the model's estimated probability for...	Individual prediction	The model may predict P(>50K) = 0.82.

```
Key distinction:

Feature importance asks:
"Which features matter to the model overall?"

LIME weight asks:
"Which feature conditions influenced THIS prediction?"

Prediction probability asks:
"How confident is the model in the predicted class?"

These three quantities should not be treated as interchangeable.
```

Figure 33: Interpretation of a positive LIME weight for education=Bachelors.

Observation: The meaning of a LIME sign is always relative to the class being explained.

Question 20

How do feature importance, LIME weight and prediction probability differ?

Answer: Feature importance is generally a global measure of how much a feature matters to the model overall. A LIME weight is a local contribution from a particular feature condition for one individual prediction. Prediction probability is the model's estimated probability for a class for that individual prediction.

Question 21

Give an IF–THEN style LIME explanation.

Answer: IF capital_gain <=0, marital_status=Never-married, capital_loss <=0, hours_per_week <=40 and education_num <=9 hold for the selected instance, THEN the combination of these local feature conditions contributes to the model predicting <=50K. This is a local model explanation and does not establish causality.

Output

```
======================================================================
IF–THEN STYLE LIME EXPLANATION
======================================================================
Predicted class: <=50K
P(>50K): 0.0000

IF the following conditions hold:
  • capital_gain <= 0.00 (LIME weight = +0.4521)
  • marital_status=Never-married (LIME weight = +0.0946)
  • capital_loss <= 0.00 (LIME weight = +0.0730)
  • hours_per_week <= 40.00 (LIME weight = +0.0722)
  • education_num <= 9.00 (LIME weight = +0.0694)

THEN the combination of these local feature conditions contributes to the model predicting <=50K.

Note: This is a local model explanation. It describes how the
model's decision is influenced by the feature conditions for this
specific individual; it does not establish causal relationships.
```

Figure 35: IF–THEN style local explanation.

Part E — Misclassification Analysis

Question 22

Which incorrectly classified test instance was selected?

Answer: The notebook identified 1,423 incorrectly classified test instances and selected test index 8. Its actual class is >50K, while the predicted class is <=50K, with P(>50K)=0.1867.

Output

```
======================================================================
LIME EXPLANATION OF MISCLASSIFICATION
======================================================================
capital_gain <= 0.00                                 +0.4459   POSITIVE – supports predicted class
marital_status=Married-civ-spouse                    -0.1126   NEGATIVE – opposes predicted class
hours_per_week <= 40.00                              +0.0721   POSITIVE – supports predicted class
education_num <= 9.00                                +0.0673   POSITIVE – supports predicted class
capital_loss <= 0.00                                 +0.0636   POSITIVE – supports predicted class
occupation=Exec-managerial                           -0.0625   NEGATIVE – opposes predicted class
relationship=Husband                                 -0.0564   NEGATIVE – opposes predicted class
sex=Male                                             -0.0277   NEGATIVE – opposes predicted class
education=HS-grad                                    +0.0275   POSITIVE – supports predicted class
race=White                                           -0.0137   NEGATIVE – opposes predicted class

TOP POSITIVE CONTRIBUTORS TO INCORRECT PREDICTION
capital_gain <= 0.00 | +0.4459
hours_per_week <= 40.00 | +0.0721
education_num <= 9.00 | +0.0673
capital_loss <= 0.00 | +0.0636
education=HS-grad | +0.0275

TOP NEGATIVE CONTRIBUTORS TO INCORRECT PREDICTION
marital_status=Married-civ-spouse | -0.1126
occupation=Exec-managerial | -0.0625
relationship=Husband | -0.0564
sex=Male | -0.0277
race=White | -0.0137
```

Figure 36: Selected misclassified instance and feature values.

Observation: The selected case is a false negative for the >50K class.

Question 23

What does LIME show for the misclassified instance?

Answer: The strongest positive contributor toward the incorrect <=50K prediction is capital_gain <=0.00 (+0.4459), followed by hours_per_week <=40.00 (+0.0721), education_num <=9.00 (+0.0673), capital_loss <=0.00 (+0.0636) and education=HS-grad (+0.0275).

Output

```
======================================================================
MODEL ERROR ANALYSIS
======================================================================
Actual class: >50K
Predicted class: <=50K
P(>50K): 0.1867
Distance from 0.50 decision boundary: 0.3133

Boundary interpretation:
The prediction is not extremely close to the decision boundary, suggesting that the model made a re

The LIME explanation can provide a possible model-level reason for
the error by showing which local feature conditions pushed the model
toward the incorrect class.

Positive contributors supported the predicted class, while negative
contributors opposed it. If strong contributions exist in both
directions, this indicates conflicting evidence in the individual's
feature profile.

LIME does not establish that these features caused the real-world
income outcome. It explains the behaviour of the trained model.
```

Figure 37: LIME explanation of the misclassified instance.

Observation: The explanation identifies the local feature conditions that pushed the model toward the wrong class.

Question 24

Which contributors support and oppose the incorrect prediction?

Answer: Positive contributors supporting <=50K are capital_gain <=0.00, hours_per_week <=40.00, education_num <=9.00, capital_loss <=0.00 and education=HS-grad. Negative contributors opposing the predicted class are married-civ-spouse (-0.1126), Exec-managerial (-0.0625), Husband (-0.0564), Male (-0.0277) and White (-0.0137).

Output

```
======================================================================
MODEL ERROR ANALYSIS
======================================================================
Actual class: >50K
Predicted class: <=50K
P(>50K): 0.1867
Distance from 0.50 decision boundary: 0.3133

Boundary interpretation:
The prediction is not extremely close to the decision boundary, suggesting that the model made a re

The LIME explanation can provide a possible model-level reason for
the error by showing which local feature conditions pushed the model
toward the incorrect class.

Positive contributors supported the predicted class, while negative
contributors opposed it. If strong contributions exist in both
directions, this indicates conflicting evidence in the individual's
feature profile.

LIME does not establish that these features caused the real-world
income outcome. It explains the behaviour of the trained model.
```

Figure 38: Positive and negative contributors for the misclassified instance.

Question 25

Why did the model likely make this error according to the LIME analysis?

Answer: The local explanation contains conflicting evidence. The strong positive contribution of capital_gain <=0.00 supports the incorrect <=50K prediction, while marital status, occupation and relationship provide evidence against that prediction. The predicted P(>50K)=0.1867 is 0.3133 away from the 0.50 decision boundary, indicating that the model was not simply making a barely-thresholded decision; it was relatively confident while still being wrong.

Output

```
Q26 — Can LIME Prove Causation?
================================

No. LIME cannot prove that a particular feature caused the model to
make an error.

MODEL ASSOCIATION:
A feature may be associated with a model prediction because the model
has learned patterns involving that feature.

FEATURE CONTRIBUTION:
A LIME weight indicates that a particular feature condition contributed
to the model's local prediction for the selected instance.

CAUSAL RELATIONSHIP:
A causal relationship means that changing the feature would actually
cause a change in the real-world outcome, under an appropriate causal
framework.

LIME only provides a local explanation of model behaviour. Therefore,
a large positive or negative LIME weight should not be interpreted as
proof of causation.

For example, if education = Bachelors has a positive LIME weight, this
means the condition contributed toward the model's prediction for that
individual. It does not prove that obtaining a Bachelor's degree would
cause the person's income to cross the income threshold.
```

Figure 39: Model error analysis and decision-boundary interpretation.

Observation: LIME helps expose conflicting local evidence and provides a model-level explanation for the observed error, but it cannot establish why the real-world income outcome occurred.

Question 26

Can LIME prove causation?

Answer: No. LIME cannot prove that a feature caused the model prediction or caused a real-world income outcome. A LIME weight indicates that a feature condition contributed to the model's local prediction. Causal claims require an appropriate causal framework and cannot be established from a local surrogate explanation alone.

Output

```
==========================
LIME STABILITY EXPERIMENT
==========================
Selected test index: 0
Actual class: <=50K
Predicted class: <=50K
P(>50K): 0.0
Run 1/10 completed.
Run 2/10 completed.
Run 3/10 completed.
Run 4/10 completed.
Run 5/10 completed.
Run 6/10 completed.
Run 7/10 completed.
Run 8/10 completed.
Run 9/10 completed.
Run 10/10 completed.
```

Figure 40: Explanation distinguishing model association, feature contribution and causal relationship.

Observation: The experiment therefore treats LIME as an explanation of model behaviour, not as a causal inference method.

Part F — LIME Stability Analysis

Question 27

How was LIME stability tested?

Answer: The same test instance, test index 0, and the same Random Forest model were explained ten times using LIME with 5,000 perturbed samples per run. The notebook completed all ten runs.

Output

TOP FIVE LIME FEATURES FOR EACH RUN

	Run	Feature 1	Feature 2	Feature 3	Feature 4	Feature 5
0	1	capital_gain <= 0.00 (+0.4421)	marital_status=Never-married (+0.0976)	hours_per_week <= 40.00 (+0.0738)	capital_loss <= 0.00 (+0.0659)	education_num <= 9.00 (+0.0638)
1	2	capital_gain <= 0.00 (+0.4272)	marital_status=Never-married (+0.0915)	hours_per_week <= 40.00 (+0.0723)	capital_loss <= 0.00 (+0.0701)	education_num <= 9.00 (+0.0632)
2	3	capital_gain <= 0.00 (+0.4519)	marital_status=Never-married (+0.0949)	hours_per_week <= 40.00 (+0.0769)	education_num <= 9.00 (+0.0682)	capital_loss <= 0.00 (+0.0641)
3	4	capital_gain <= 0.00 (+0.4525)	marital_status=Never-married (+0.0902)	capital_loss <= 0.00 (+0.0784)	hours_per_week <= 40.00 (+0.0749)	education_num <= 9.00 (+0.0606)
4	5	capital_gain <= 0.00 (+0.4373)	marital_status=Never-married (+0.0975)	hours_per_week <= 40.00 (+0.0723)	education_num <= 9.00 (+0.0703)	capital_loss <= 0.00 (+0.0628)
5	6	capital_gain <= 0.00 (+0.4337)	marital_status=Never-married (+0.0926)	capital_loss <= 0.00 (+0.0821)	hours_per_week <= 40.00 (+0.0753)	education_num <= 9.00 (+0.0746)
6	7	capital_gain <= 0.00 (+0.4440)	marital_status=Never-married (+0.0892)	hours_per_week <= 40.00 (+0.0745)	education_num <= 9.00 (+0.0729)	capital_loss <= 0.00 (+0.0558)
7	8	capital_gain <= 0.00 (+0.4533)	marital_status=Never-married (+0.0947)	hours_per_week <= 40.00 (+0.0733)	capital_loss <= 0.00 (+0.0686)	education_num <= 9.00 (+0.0658)
8	9	capital_gain <= 0.00 (+0.4282)	marital_status=Never-married (+0.0931)	capital_loss <= 0.00 (+0.0762)	hours_per_week <= 40.00 (+0.0751)	education_num <= 9.00 (+0.0650)
9	10	capital_gain <= 0.00 (+0.4342)	marital_status=Never-married (+0.0979)	education_num <= 9.00 (+0.0720)	hours_per_week <= 40.00 (+0.0701)	capital_loss <= 0.00 (+0.0694)

Figure 41: Ten-run LIME stability experiment completion.

Question 28

What were the top five features across the ten runs?

Answer: The five conditions that appeared in the top five in all ten runs were capital_gain <=0.00, marital_status=Never-married, hours_per_week <=40.00, capital_loss <=0.00 and education_num <=9.00. The ordering of some lower positions varied slightly between runs.

Output

LIME STABILITY SUMMARY

	Mean LIME Weight	Standard Deviation	Minimum LIME Weight	Maximum LIME Weight	Top-5 Appearances	CV (%)
capital_gain <= 0.00	0.4404	0.0099	0.4272	0.4533	10	2.2420
marital_status=Never-married	0.0939	0.0031	0.0892	0.0979	10	3.3349
hours_per_week <= 40.00	0.0739	0.0019	0.0701	0.0769	10	2.6236
capital_loss <= 0.00	0.0693	0.0079	0.0558	0.0821	10	11.3641
education_num <= 9.00	0.0676	0.0047	0.0606	0.0746	10	6.9288
relationship=Own-child	0.0458	0.0024	0.0429	0.0504	0	5.3186
occupation=Machine-op-inspct	0.0290	0.0053	0.0212	0.0364	0	18.1335
sex=Male	-0.0283	0.0035	-0.0322	-0.0222	0	12.4503
education=HS-grad	0.0267	0.0030	0.0222	0.0312	0	11.3831
race=White	-0.0072	0.0066	-0.0154	0.0000	0	91.1905
workclass=Private	0.0027	0.0057	0.0000	0.0154	0	213.6541
fnlwgt > 238329.00	-0.0016	0.0052	-0.0163	0.0000	0	316.2278
native_country=United-States	-0.0013	0.0040	-0.0127	0.0000	0	316.2278

Figure 42: Top five LIME features recorded for each of the ten runs.

Observation: The repeated appearance of the same five conditions indicates strong rank stability for the main local explanation.

Question 29

What were the stability statistics?

Answer: The notebook calculated mean LIME weight, standard deviation, minimum, maximum, top-five appearances and coefficient of variation for the observed conditions.

Feature condition	Mean	SD	Min	Max	Top-5	CV (%)
capital_gain <= 0.00	0.4404	0.0099	0.4272	0.4533	10	2.2420
marital_status=Never-married	0.0939	0.0031	0.0892	0.0979	10	3.3349
hours_per_week <= 40.00	0.0739	0.0019	0.0701	0.0769	10	2.6236
capital_loss <= 0.00	0.0693	0.0079	0.0558	0.0821	10	11.3641
education_num <= 9.00	0.0676	0.0047	0.0606	0.0746	10	6.9288
relationship=Own-child	0.0458	0.0024	0.0429	0.0504	0	5.3186
occupation=Machine-op-inspct	0.0290	0.0053	0.0212	0.0364	0	18.1335
sex=Male	-0.0283	0.0035	-0.0322	-0.0222	0	12.4503
education=HS-grad	0.0267	0.0030	0.0222	0.0312	0	11.3831
race=White	-0.0072	0.0066	-0.0154	0.0000	0	91.1905
workclass=Private	0.0027	0.0057	0.0000	0.0154	0	213.6541
fnlwgt > 238329.00	-0.0016	0.0052	-0.0163	0.0000	0	316.2278
native_country=United-States	-0.0013	0.0040	-0.0127	0.0000	0	316.2278

Output

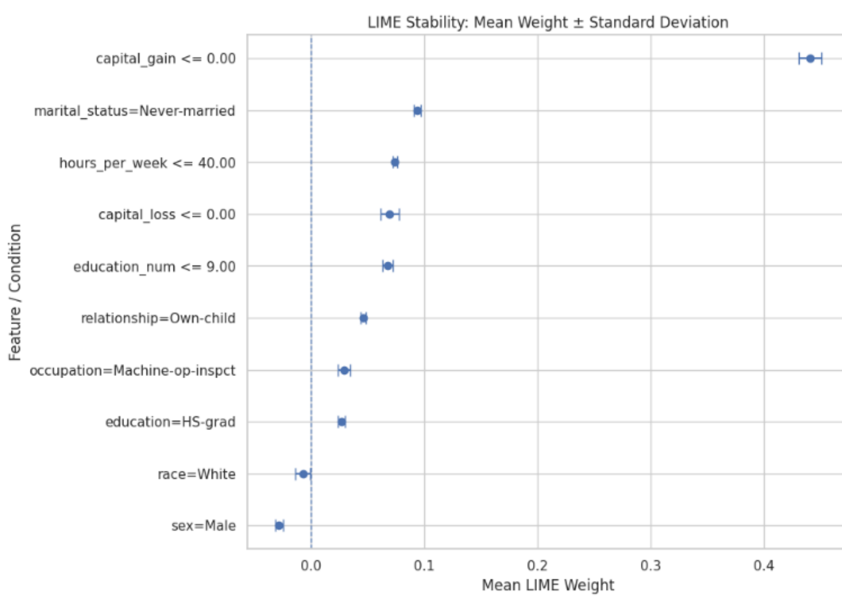


Figure 43: LIME stability summary table.

Observation: The five main conditions all appeared in the top five in every run. capital_gain <=0.00 had the highest mean contribution (0.4404), while hours_per_week <=40.00 had the smallest standard deviation among the consistently top-five conditions (0.0019).

Question 30

How was LIME stability visualized?

Answer: The notebook plotted the mean LIME weight for the most important conditions with error bars representing standard deviation. Smaller error bars indicate that the local contribution changed less across repeated runs.

Output

```
============================================================
STABLE AND UNSTABLE FEATURES
============================================================

1. Most stable feature:
hours_per_week <= 40.00 | SD = 0.0019

2. Least stable feature:
capital_gain <= 0.00 | SD = 0.0099

3. Feature with highest average contribution:
capital_gain <= 0.00 | Mean = 0.4404

4. Feature with largest variation:
capital_gain <= 0.00 | SD = 0.0099
```

Figure 44: Mean LIME weight with standard-deviation error bars.

Observation: The visualisation makes both contribution magnitude and variability easy to compare.

Question 31

Which features were most and least stable?

Answer: Among the consistently top-five conditions, hours_per_week <=40.00 was the most stable with SD=0.0019. capital_gain <=0.00 was the least stable with SD=0.0099. capital_gain <=0.00 also had the highest average contribution, with mean=0.4404, and the largest variation in absolute standard-deviation terms among the consistently top-five conditions.

Output

```
Q32 — Why can LIME explanations vary between runs?
==================================================

1. RANDOM PERTURBATIONS
LIME generates perturbed samples around the selected instance.
Different random samples can lead to slightly different local
approximations.

2. NUMBER OF SAMPLES
A small number of perturbed samples provides less information about
the local model behaviour. Increasing num_samples generally gives
LIME more information from the local neighbourhood.

3. LOCAL NEIGHBOURHOOD
LIME focuses on observations near the instance being explained.
The characteristics of this neighbourhood influence the resulting
surrogate model.

4. KERNEL WIDTH
LIME gives different weights to perturbed observations according to
their proximity to the original instance. The kernel width controls
how quickly that influence decreases with distance.

5. MODEL COMPLEXITY
A complex model such as Random Forest can have nonlinear and
interaction effects. A simple local surrogate may approximate these
patterns differently depending on the sampled neighbourhood.

Therefore, some variation between LIME runs is expected. A stable
feature should generally maintain a similar contribution across
different runs, while unstable features may have large changes in
weight or may appear in the top features inconsistently.
```

Figure 45: Stable and unstable feature summary.

Observation: A feature can be both highly influential and somewhat variable. Therefore, contribution magnitude and stability should be considered separately.

Question 32

Why can LIME explanations vary between runs?

Answer: LIME relies on random perturbations around the selected instance. Different perturbed samples can lead to slightly different local surrogate models. Variation can also arise from the number of samples, the local neighbourhood, kernel width and the nonlinear or interaction behaviour of the Random Forest. Therefore, some variation between runs is expected, and stability should be assessed when an explanation is used for an important decision.

Output

```
Running stability experiment for num_samples = 1000
  Repeat 1/5 complete
  Repeat 2/5 complete
  Repeat 3/5 complete
  Repeat 4/5 complete
  Repeat 5/5 complete

Running stability experiment for num_samples = 5000
  Repeat 1/5 complete
  Repeat 2/5 complete
  Repeat 3/5 complete
  Repeat 4/5 complete
  Repeat 5/5 complete

Running stability experiment for num_samples = 10000
  Repeat 1/5 complete
  Repeat 2/5 complete
  Repeat 3/5 complete
  Repeat 4/5 complete
  Repeat 5/5 complete

======================================================================
COMPARISON OF DIFFERENT num_samples VALUES
======================================================================
```

	num_samples	Average Standard Deviation	Maximum Standard Deviation	Average Absolute LIME Weight
0	1000	0.0094	0.0200	0.0721
1	5000	0.0052	0.0090	0.0731
2	10000	0.0037	0.0079	0.0737

Figure 46: Reasons for variation between repeated LIME runs.

Question 33

What was repeated for different num_samples values?

Answer: The stability experiment was repeated five times for each of three LIME sample sizes: 1,000, 5,000 and 10,000 perturbation samples.

Output

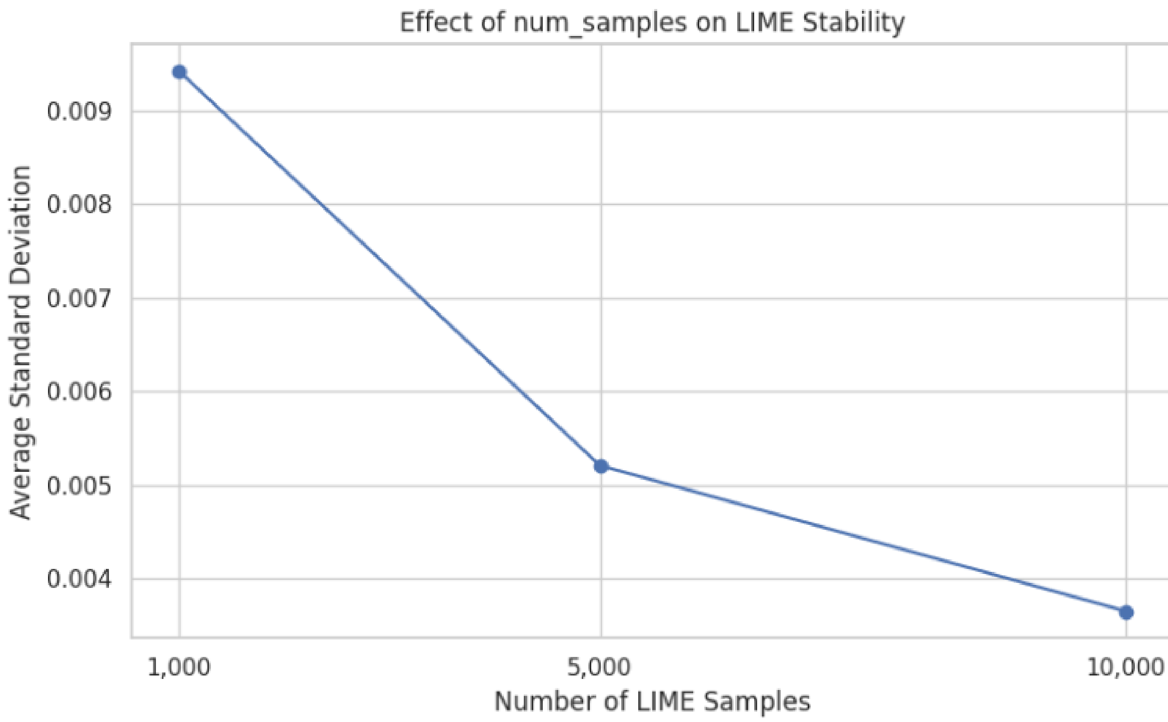


Figure 47: Repeated stability experiments for 1,000, 5,000 and 10,000 samples and comparison table.

Question 34

How did num_samples affect LIME stability?

Answer: Increasing the number of perturbed samples reduced explanation variability. The average standard deviation decreased from 0.0094 at 1,000 samples to 0.0052 at 5,000 and 0.0037 at 10,000. The maximum standard deviation decreased from 0.0200 to 0.0090 and then 0.0079. The average absolute LIME weight remained similar, changing from 0.0721 to 0.0731 and 0.0737.

Output

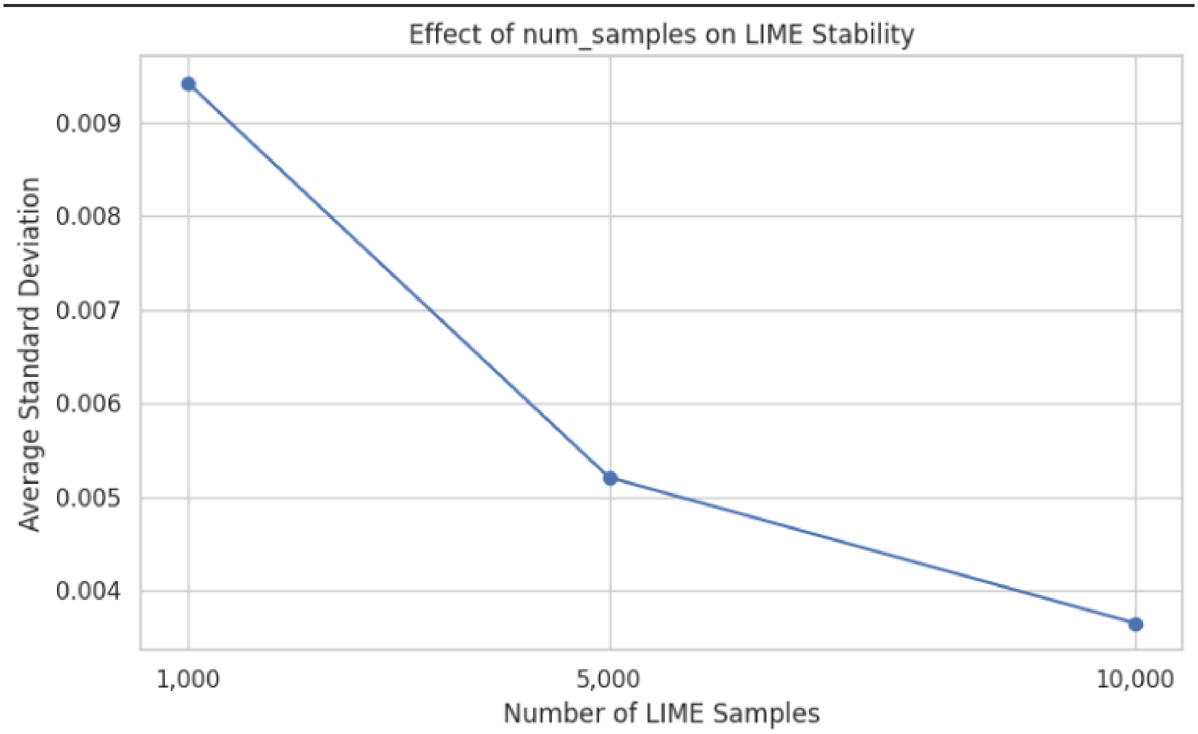


Figure 48: Effect of num_samples on average LIME standard deviation.

Observation: The decreasing standard deviation shows that larger perturbation samples produced more stable local explanations in this experiment. The average absolute LIME weight changed only slightly, indicating that increasing sample size mainly improved stability rather than changing the overall magnitude of contributions.

Comparative Analysis

The model evaluation and explainability results show that predictive performance and interpretability are different dimensions. Random Forest achieved 85.43% accuracy and ROC-AUC 0.9044, while Logistic Regression achieved 80.79% accuracy and the same ROC-AUC. Random Forest was selected for LIME because its ensemble structure is harder to inspect directly.

The LIME explanations show that local feature contributions are not fixed global effects. For example, capital_gain <=0.00 had a strong positive contribution toward the <=50K class for Instances 1 and 3, but a strong negative contribution when the >50K class was being predicted for Instance 2. This demonstrates why LIME explanations must always be interpreted in the context of the selected individual and explained class.

The misclassification analysis further shows the value of local explanations. In the selected error, the model received strong evidence in both directions. LIME identifies the feature conditions that supported the incorrect class and the conditions that opposed it, making the model's behaviour easier to inspect than an unexplained prediction alone.

The stability analysis also demonstrates that LIME is stochastic. Although the exact weights vary, the five strongest conditions remained in the top five across all ten 5,000-sample runs. Increasing num_samples reduced both average and maximum standard deviation, indicating that more perturbation samples improved stability for this experiment.

Conclusion

This laboratory successfully applied Local Interpretable Model-Agnostic Explanations (LIME) to a tabular Adult/Census Income classification problem. The experiment covered dataset understanding, data-quality analysis, preprocessing, Random Forest and Logistic Regression training, model evaluation, black-box model selection, local explanations, misclassification analysis, causal limitations and stability testing.

The results show that LIME can make an individual Random Forest prediction easier to inspect by identifying feature conditions that support or oppose the class being explained. However, LIME weights are local model contributions rather than global feature importance measures and do not establish causal relationships. The selected misclassification demonstrated that a model can be relatively confident while still receiving conflicting local evidence.

The stability experiment found that the main top-five conditions remained consistent across repeated runs. Increasing num_samples from 1,000 to 10,000 reduced average standard deviation from 0.0094 to 0.0037 and maximum standard deviation from 0.0200 to 0.0079. Thus, in this experiment, increasing the number of perturbations produced more stable LIME explanations.

Overall, the laboratory demonstrates that explainability should be evaluated alongside predictive performance. LIME is particularly useful for inspecting individual predictions of complex tabular models, but its stochastic and local nature must be considered when interpreting or communicating the resulting explanations.

Key Findings at a Glance

- Dataset: 48,842 instances and 14 input features.
- Feature composition: 6 numerical and 8 categorical features.
- Class distribution: 76.07% <=50K and 23.93% >50K.
- Random Forest accuracy: 85.43%; ROC-AUC: 0.9044.
- Logistic Regression accuracy: 80.79%; ROC-AUC: 0.9044.
- Random Forest selected as the black-box model for LIME.
- Selected misclassification: test index 8, actual >50K, predicted <=50K, P(>50K)=0.1867.
- capital_gain <=0.00 was the strongest local contributor in several explanations.
- hours_per_week <=40.00 had the lowest standard deviation among the consistently top-five conditions.
- Average LIME standard deviation decreased from 0.0094 at 1,000 samples to 0.0037 at 10,000 samples.
- LIME explains model behaviour locally and does not prove causation.

GitHub Link

https://github.com/Aviksha-RVU3016/